

Descriptive Analytics of NHS using Python



GEORGE BREESE



The NHS is a public healthcare system in UK

- With the population rising and dealing with the aftereffects of the COVID-19 pandemic, the resources of the NHS have to be assessed. The following review will analyse two main questions posed by the NHS:
- Has there been adequate staff and capacity in the networks?
- What was the actual utilisation of resources?

This review will look only at NHS England, rather than the wider United Kingdom

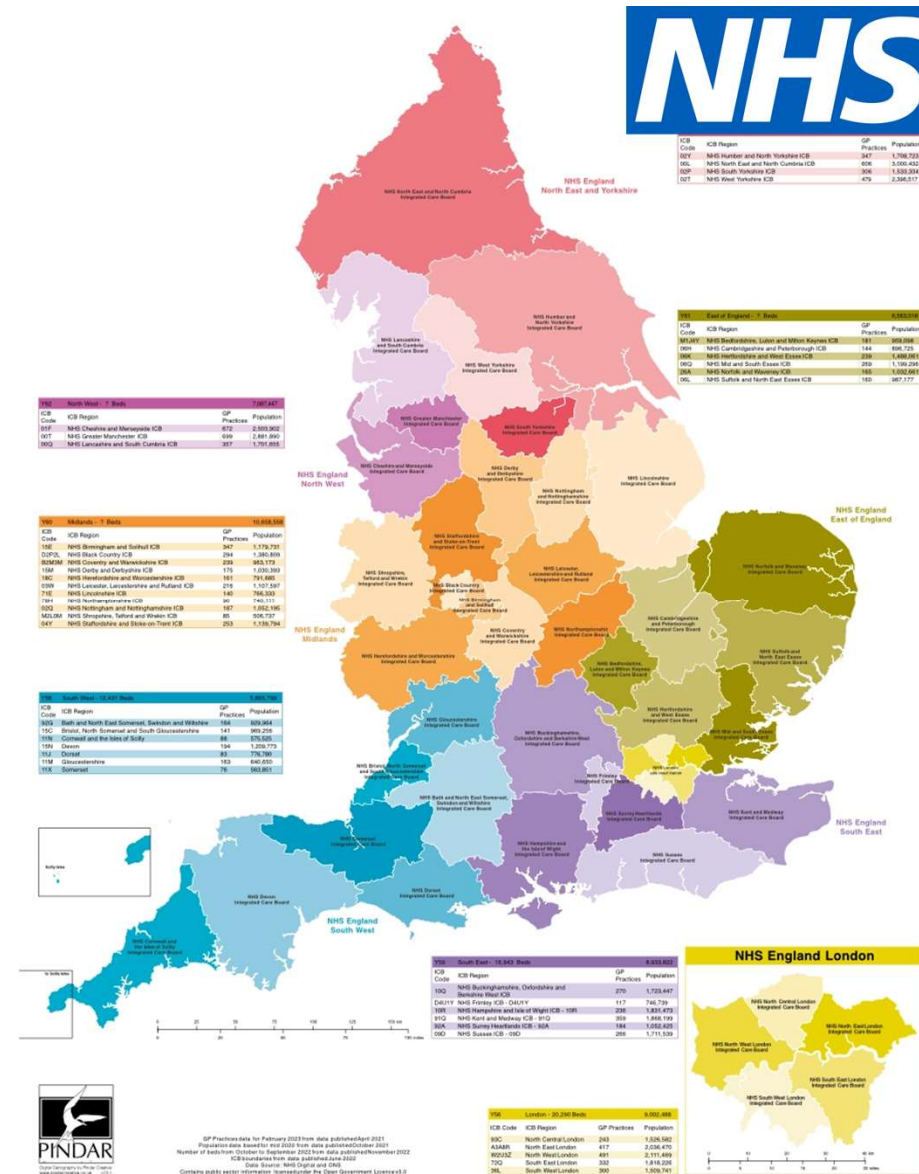
To complete the analysis, more analytical questions are asked of the data:

- How well utilised are staff compared to each month?
- Is there any link between missed appointments and the time between a booking an appointment and the appointment taking place?
- Was there adequate staff and capacity in the networks?

The insights gained from this analysis will help determine whether the current capacity of the NHS is adequate or it needs to be increased.

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- There are 42 ICBs
- There are 106 sub-ICBs



What is the number of service settings in the data sets?



- There are 5 different service settings
 - General practice
 - Primary care network
 - Other
 - Extended access provision
 - Unmapped
-

What is the number of context types in the data sets?



- There are three separate context types
 - Care related encounter
 - Inconsistent mapping
 - Unmapped
-

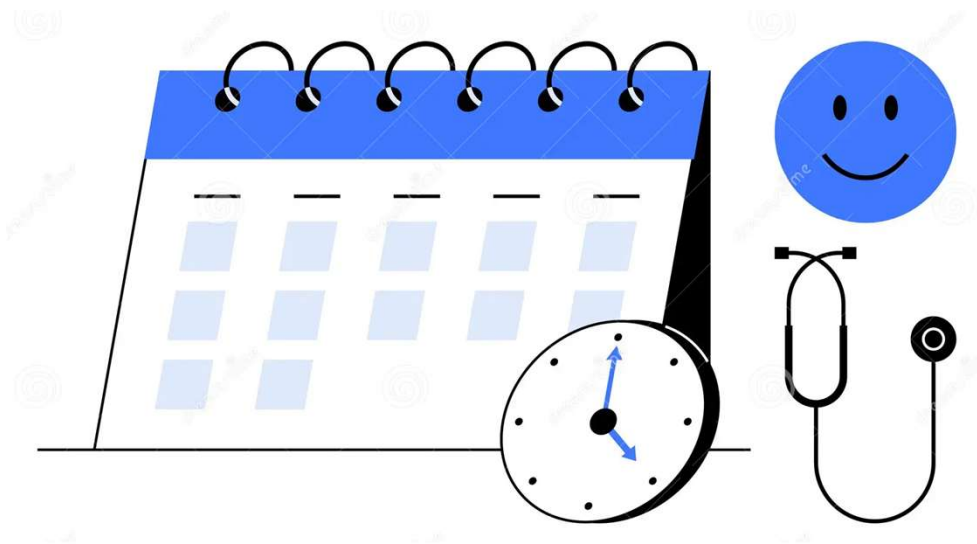
What is the number of national categories in the data sets?



- There are 18 different national categories
- A sample of them are as follows
 - General consultation routine
 - General consultation acute
 - Planned clinics
 - Clinical Triage
 - Planned clinical procedure
 - Structured medication review
 - Inconsistent mapping
 - unmapped

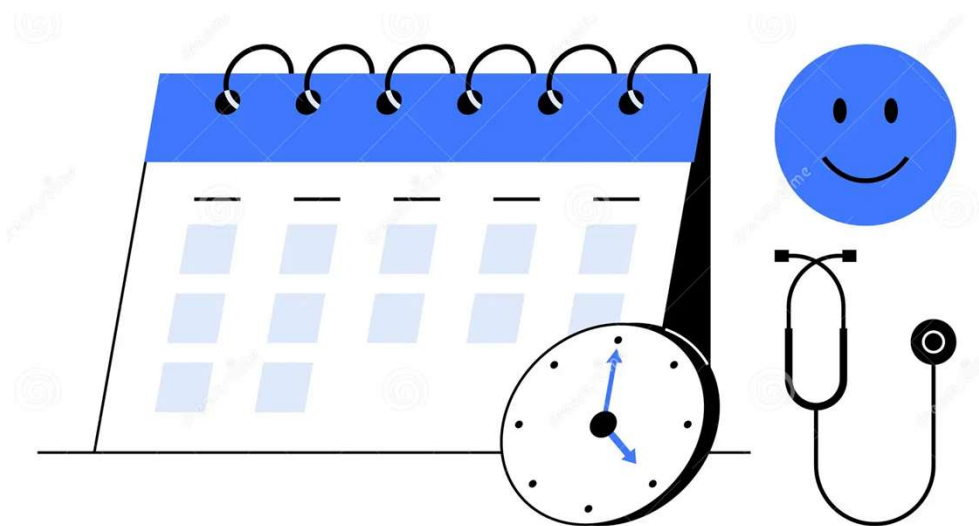


What is the number of appointment statuses in the data sets?



- There are three different appointment statuses
 - Attended
 - Unknown
 - DNA (did not attend)

What is the time period for the data?



- National categories data set
 - 1st August 2021 – 30th June 2022
- Actual duration data set
 - 1st December 2021 – 30th June 2022
- Appointments regional data set
 - January 2020 – June 2022

For this analysis we will mainly be focusing between August 2021 – June 2022.

FREE HOME DELIVERY FOR SIX WEEKS

DAILY EXPRESS

BIRTHDAY BOY ARCHIE ALL SMILES FROM LA

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FIRST STEPS TO FREEDOM FROM MONDAY

Message to 'stay at home' will be axed

UK's 30,076 death toll highest in Europe

DAILY Mirror

ADELE

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FIVE STEPS TO END OF LOCKDOWN

Staggered easing plan revealed... from Monday till October

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THURSDAY, MAY 7, 2020

Torment of a Branson betrayed

femail magazine exclusive

Billionaire's sister on her husband's shattering affair SEE PAGE 37

HURRAH! LOCKDOWN FREEDOM BECKONS

'Stay home' slogan and once-a-day exercise rule scrapped ++ Picnics, park trips and countryside outings back on

Hello, who's got a new look for lockdown?

DRACONIAN rules imposed under the six-week lockdown will start to be eased on Monday.

By Jason Groves Political Editor

announced on Sunday after the Cabinet finalises the details. The Prime Minister will also host a Covid-19 emergency meeting with leaders in Scotland, Wales and Northern Ireland in the hope of averting a UK-wide economic

SET JUST £1 FOR YOUR CHANCE TO WIN

THE Sun

ARCHIE ANGEL

INSIDE FREE PUZZLE PULLOUT

LOCKDOWN JOY NEXT WEEK

HAPPY MONDAY

- Go out to exercise all you like
- Sit in park 2m from your pals
- Pubs, cafes plan to open gardens

DAILY STAR

MAY 11

MAGIC MONDAY

Easing of lockdown could start in 4 days

Govt urges: Buy a paper (and we agree!)

Fakeaway Slimming World

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£5 OFF
WHEN YOU
SPEND £15
ON HEALTH
CARE AT...
Boots
SEE PAGE 22
Terms & exclusions apply



FREE PULLOUT
You CAN keep
your mind
sharp... and
beat dementia
with Dr SANJAY GUPTA

Letters inviting over-70s for Covid jab sent out TODAY

HOPE IN THE POST FOR 5M

By Daniel Martin
and Colin Fernandez

OVER 5million more Britons will be invited to receive a coronavirus jab from today. In a major milestone for the vaccination programme, letters will start being sent to those in England in the next two priority groups. This includes 4.6million in their 70s plus another one million classed as 'clinically extremely vulnerable' because they have conditions affecting the immune system, certain cancers or are organ transplant recipients. Health Secretary Matt Hancock said the nation was 'nearly on the home straight' as 50 per cent of all over-80s in England have been vaccinated. Some 140 a minute are receiving a jab, putting Britain on course to vaccinate all adults by early autumn. Foreign Secretary Dominic Raab pledged yesterday that every over-18 will be offered a first jab by September - if not earlier. And he said he was hopeful some lockdown measures could start to be lifted.



Outrage as
ex-top judge
tells cancer
patient, 39:
YOUR LIFE IS
UNVALUABLE



ROYAL FAMILY EXEMPT
FROM 'FEUDAL' LAND
LAW REFORM
EXCLUSIVE
SEE PAGES B&9



THE IMMUNITY DIET
Eat your
way to
good
health
expressyourself • SEE PAGES 30&31

VACCINES FOR OVER-70s START TODAY

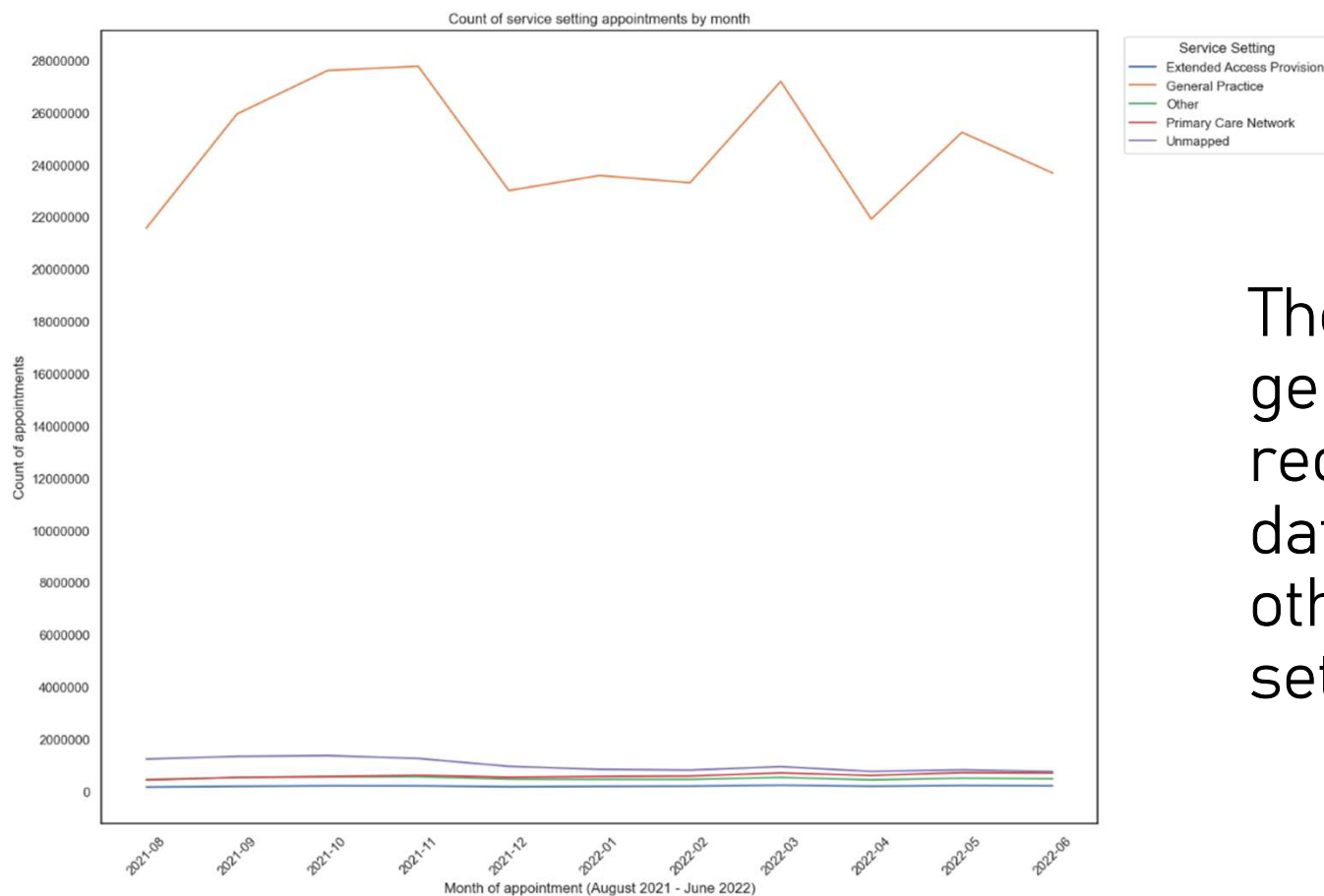
By Martyn Brown
Senior Political Correspondent

OVER-70s will be offered their Covid jabs from today as the UK vaccination rollout accelerates. Millions more clinically vulnerable shielders could now also get their call-up. Boris Johnson said 140 injections are being administered a minute - and that will rise with today's opening of 10 new hubs. All adults will be offered the vaccine by autumn, insiders said. The Prime Minister

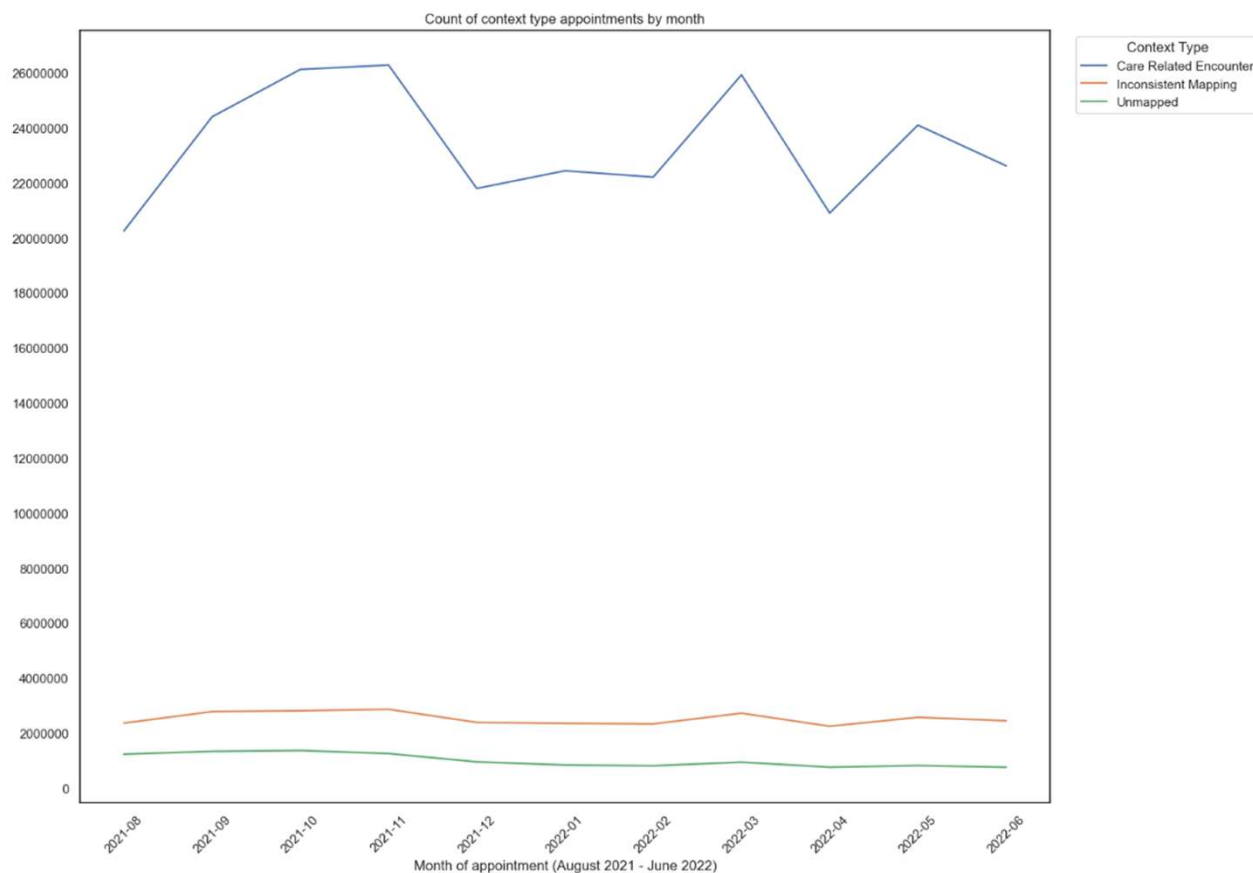


CLUMSY ON ICE
Denise battles
on through
the pain
SEE PAGE 3

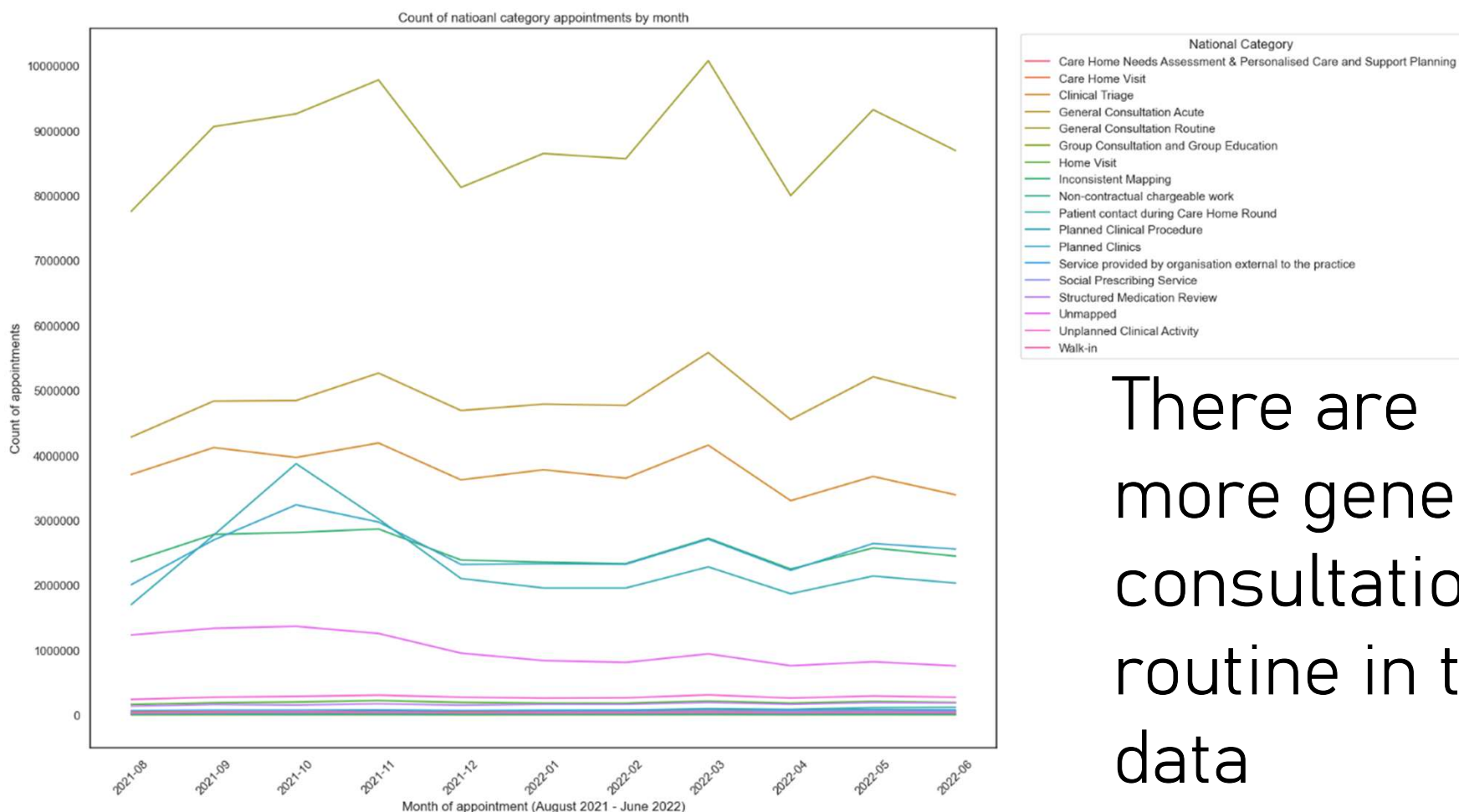
'Huge progress' says PM with 140...



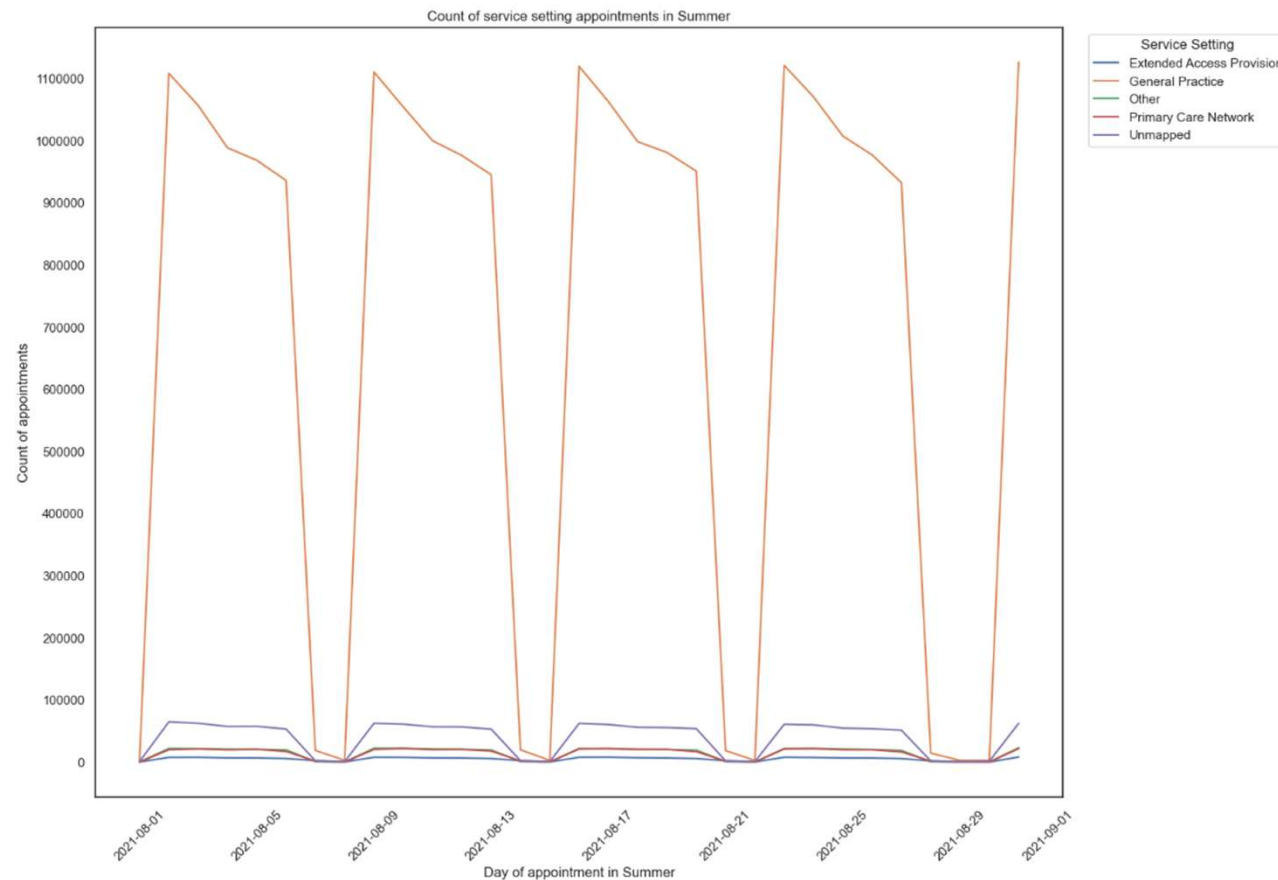
There are more general practice records in the data compared to other service settings



There are more care related encounters in the data



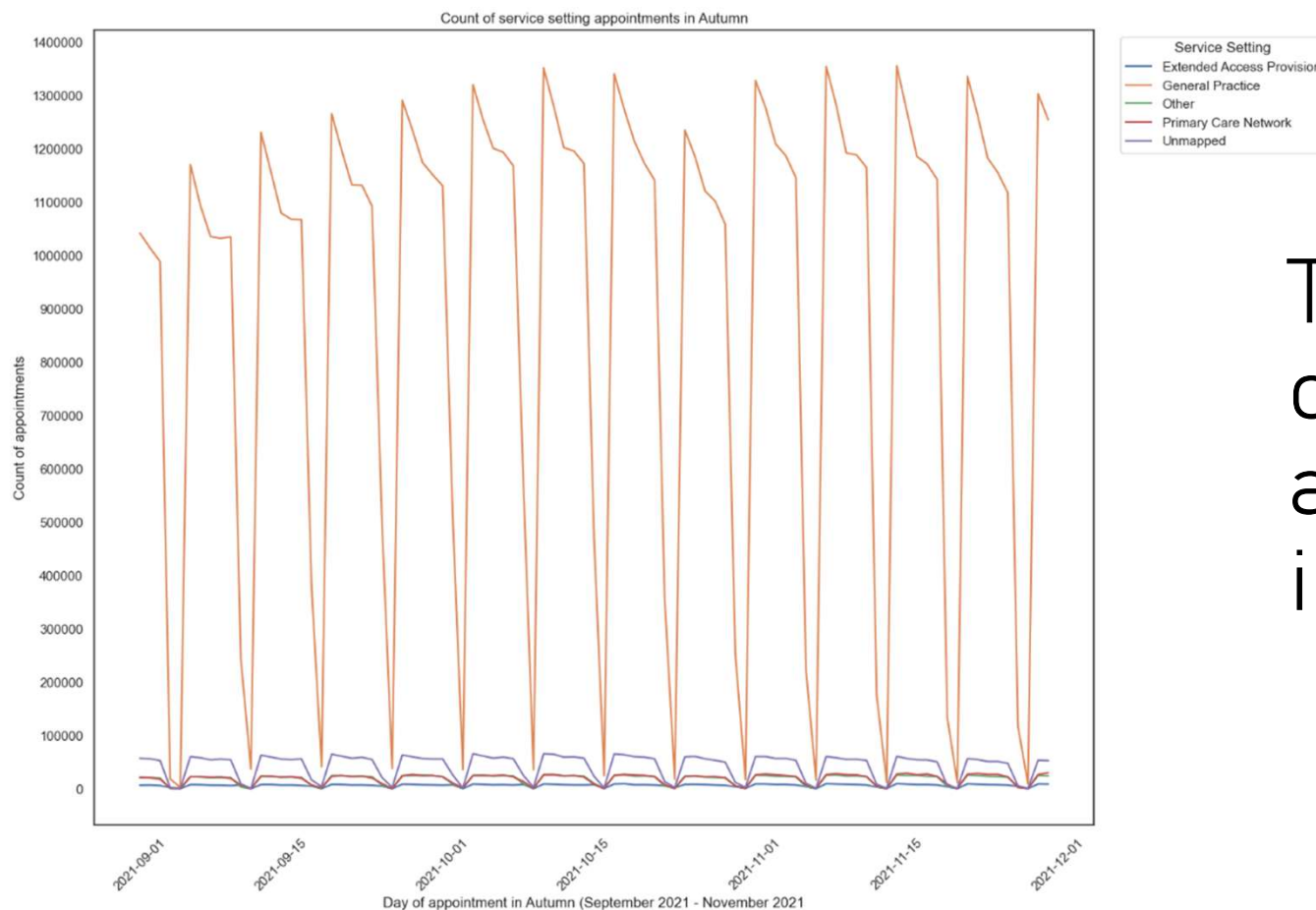
There are more general consultation routine in the data



The number of appointments at the start of the specified time in August

There are no appointments at the weekend.

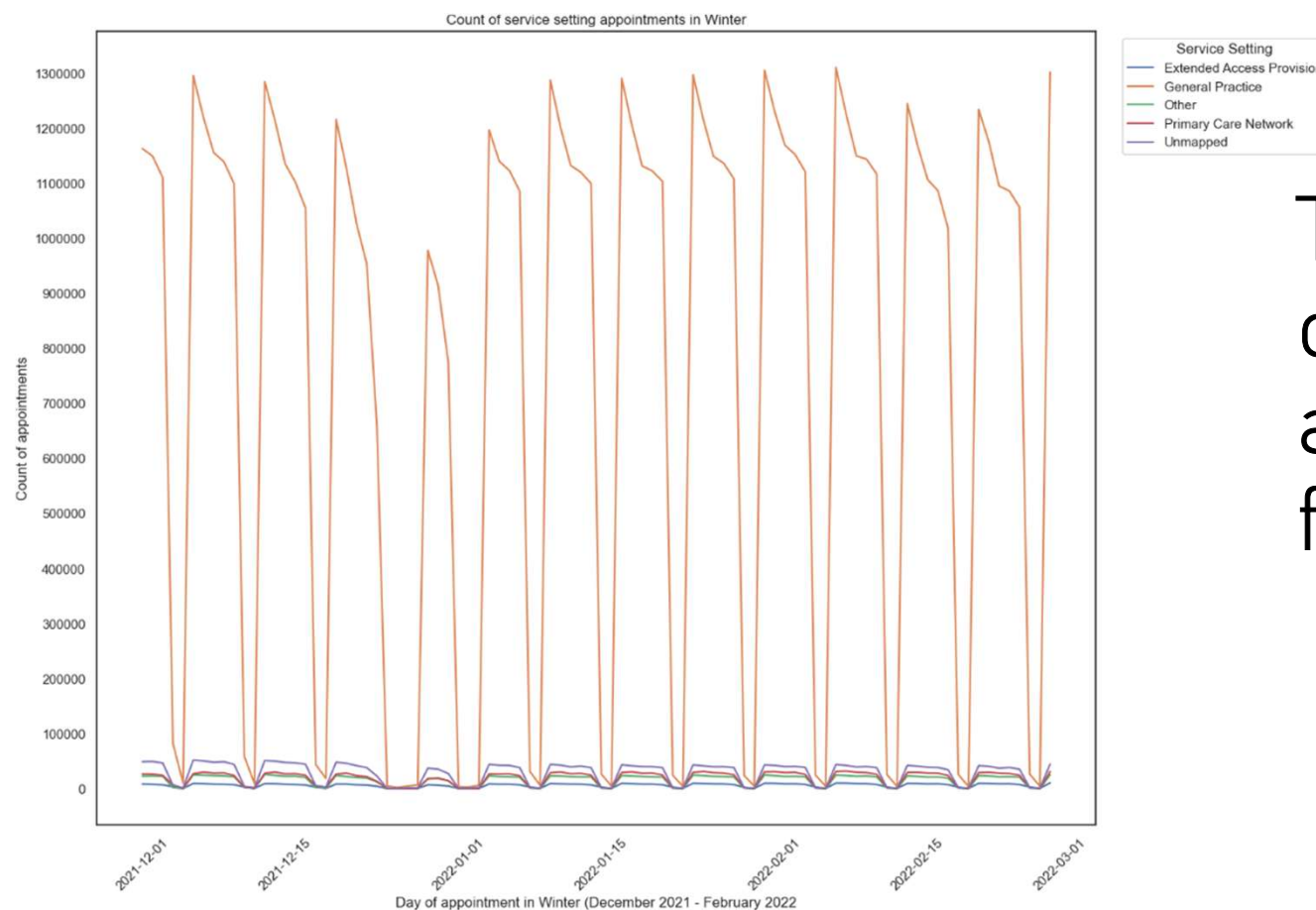
There are more appointments on a Monday compared to a Friday.



The number of appointments in Autumn

There are no appointments at the weekend.

There are more appointments on a Monday compared to a Friday.



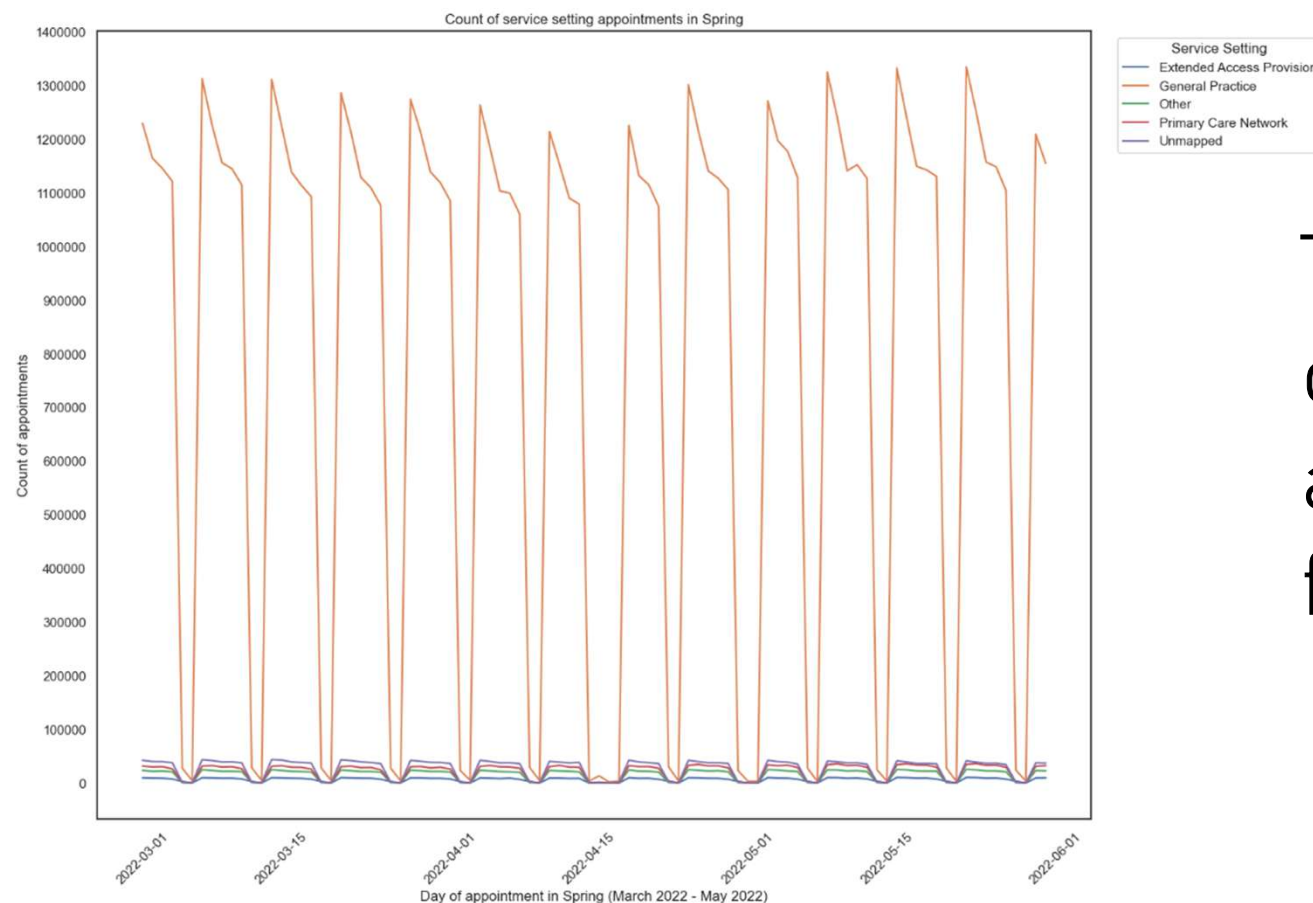
The number of appointments for Winter

There are no appointments at the weekend.

There are no appointments on bank holidays.

There are fewer appointments between Christmas and new year

There are more appointments on a Monday compared to a Friday.

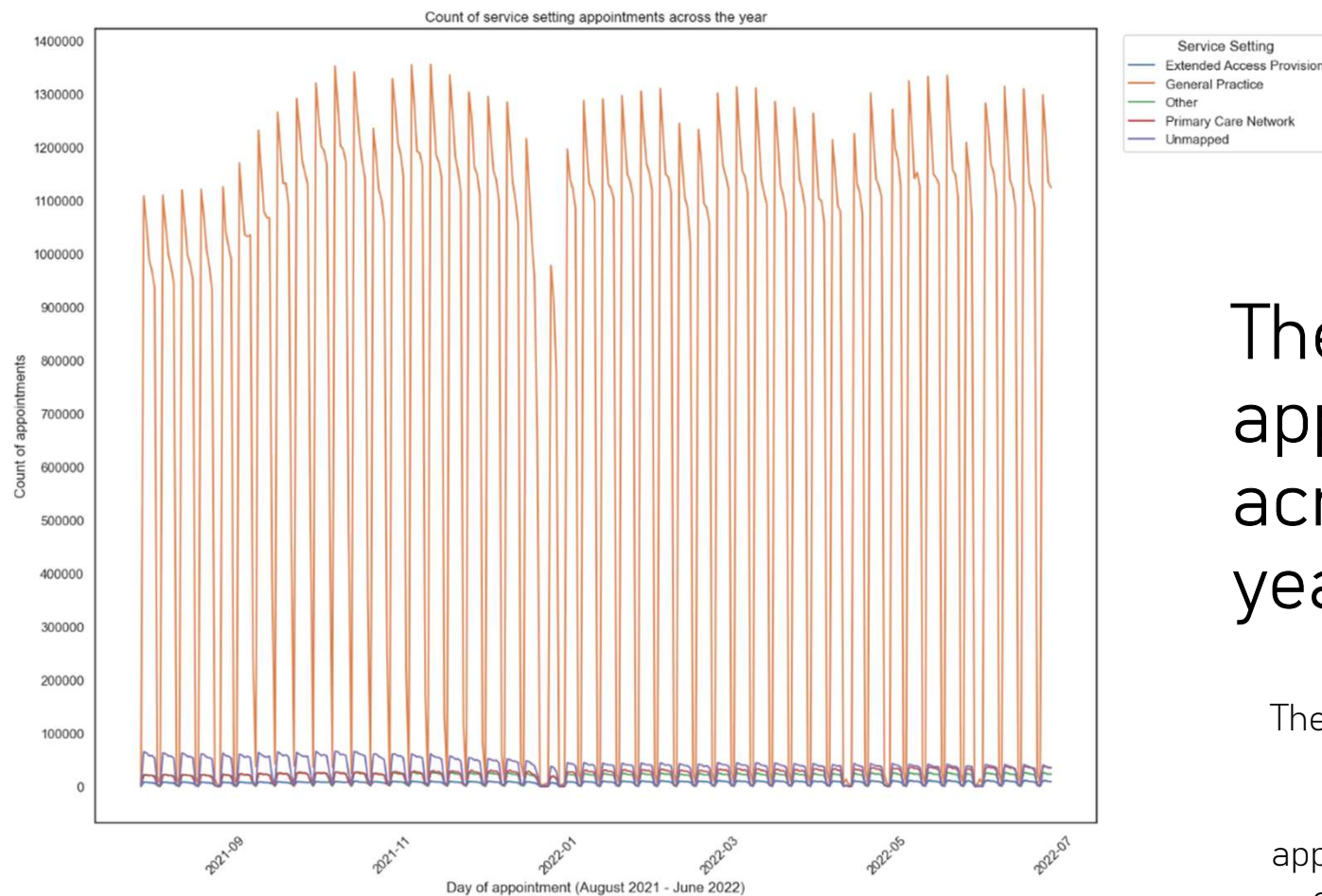


The number of appointments for Spring

There are no appointments at the weekend.

There are no appointments during bank holidays.

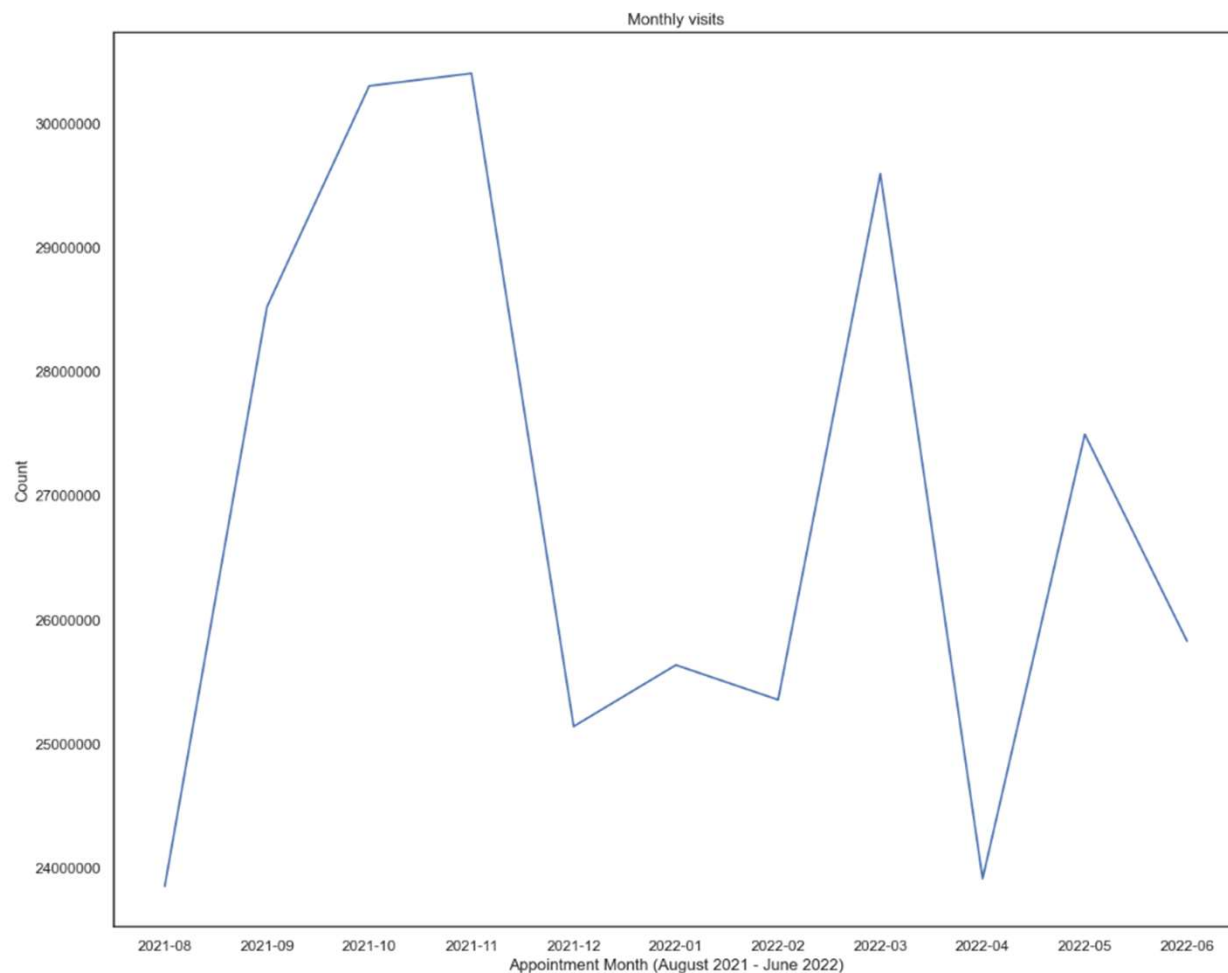
There are more appointments on a Monday compared to a Friday.



The number of appointments across the year

There are no appointments at the weekend.

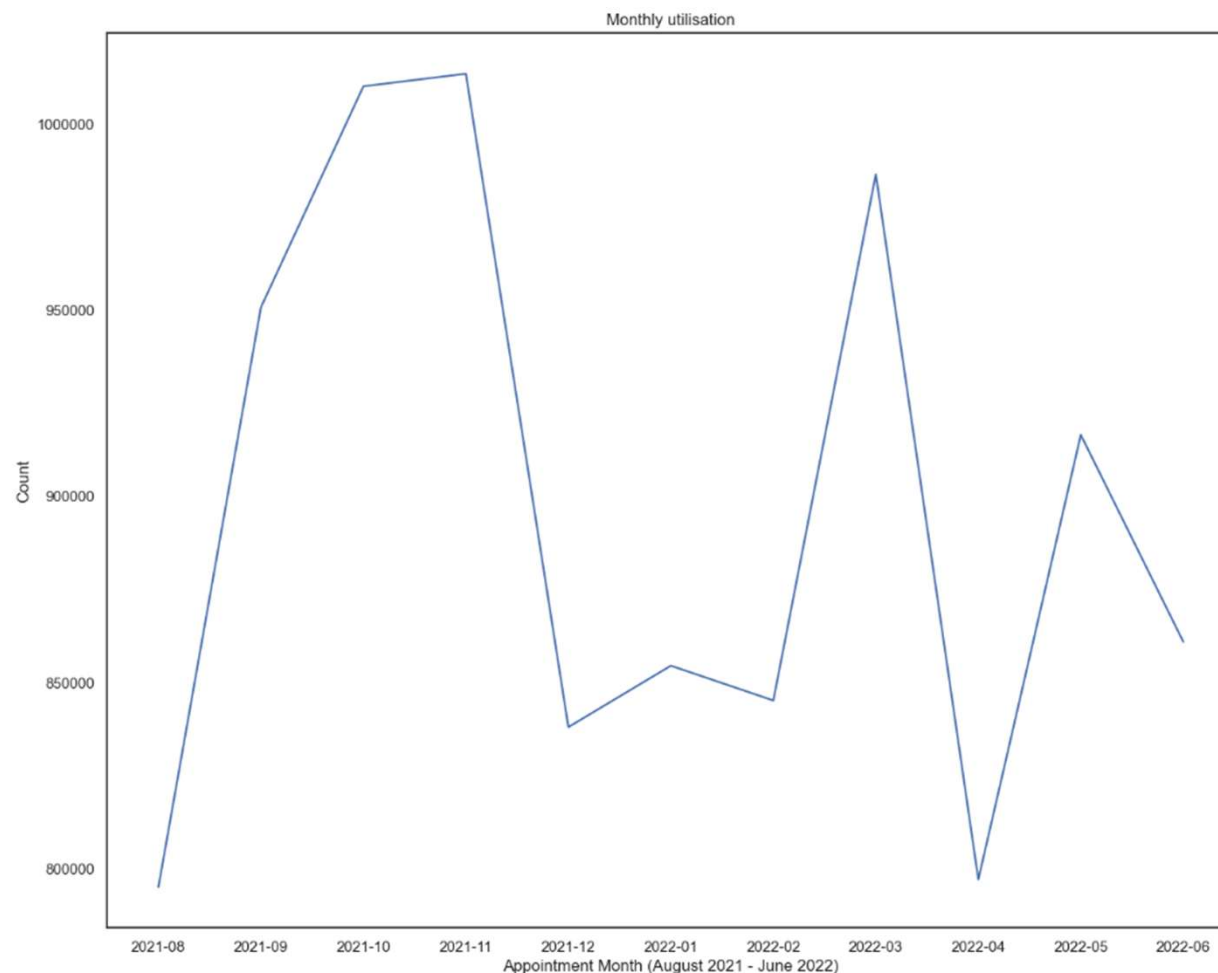
There are more appointments on a Monday compared to a Friday.

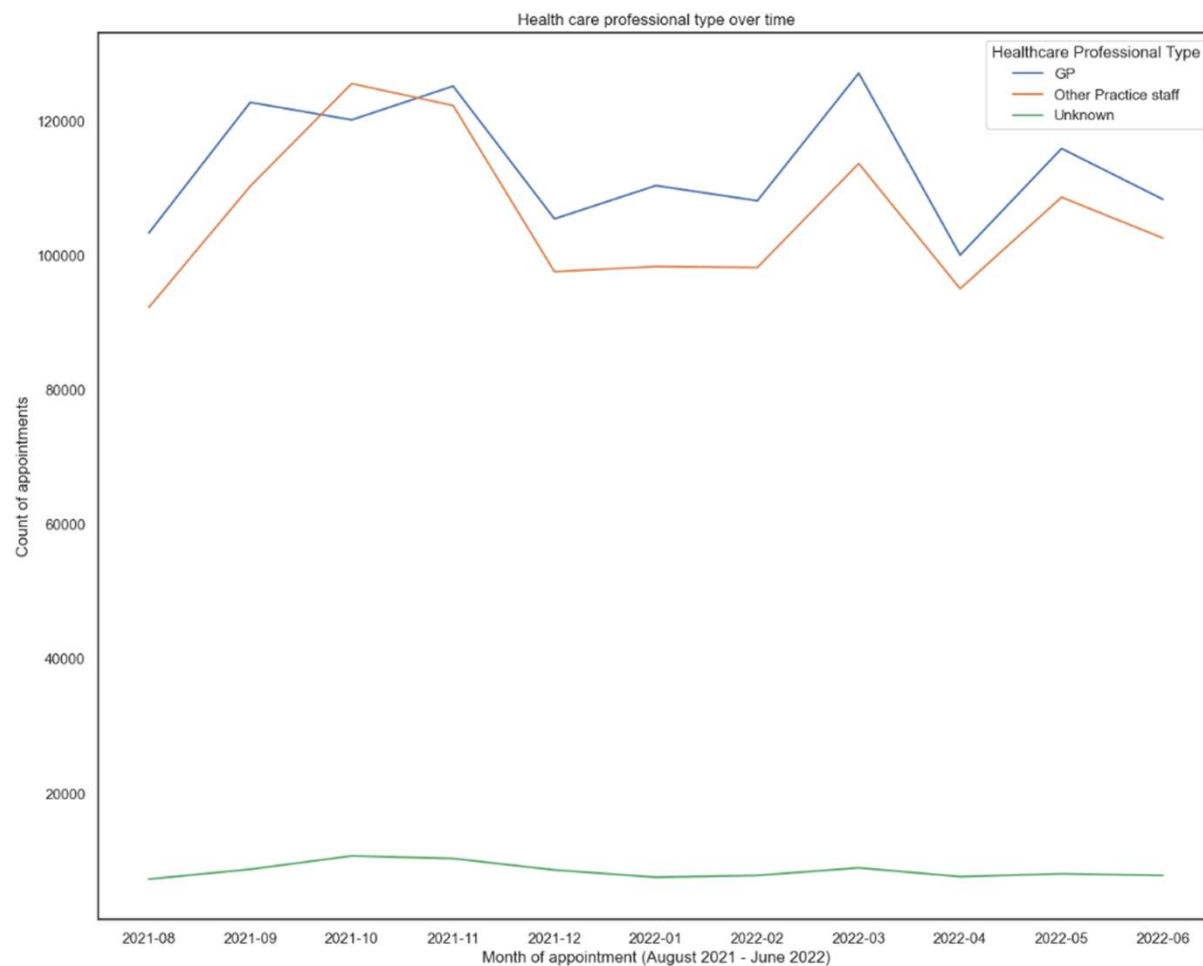


Count of
appointments
compared to
the month

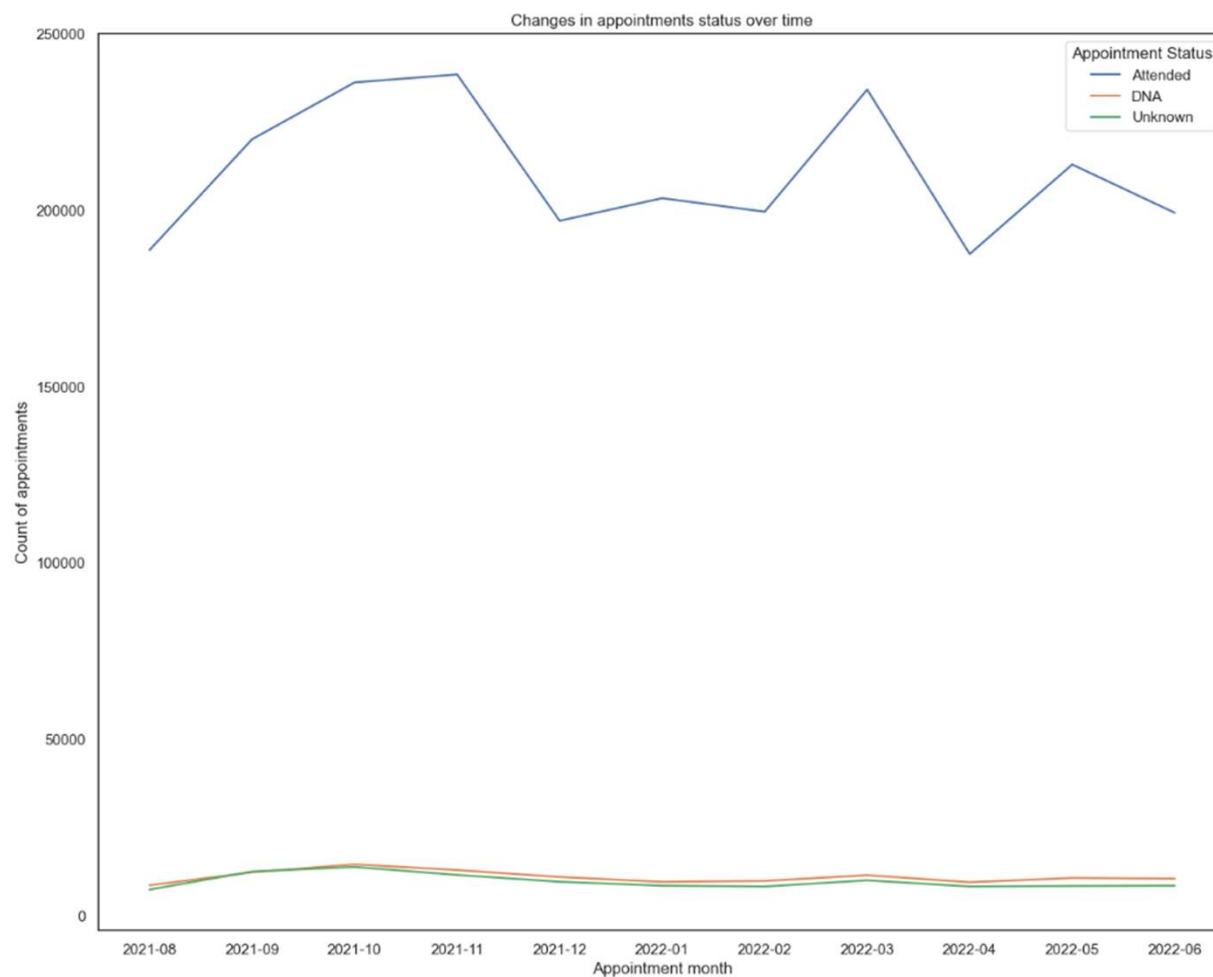
Average utilisation compared to the month

The NHS has a capacity of 120000 appointment to utilise per month

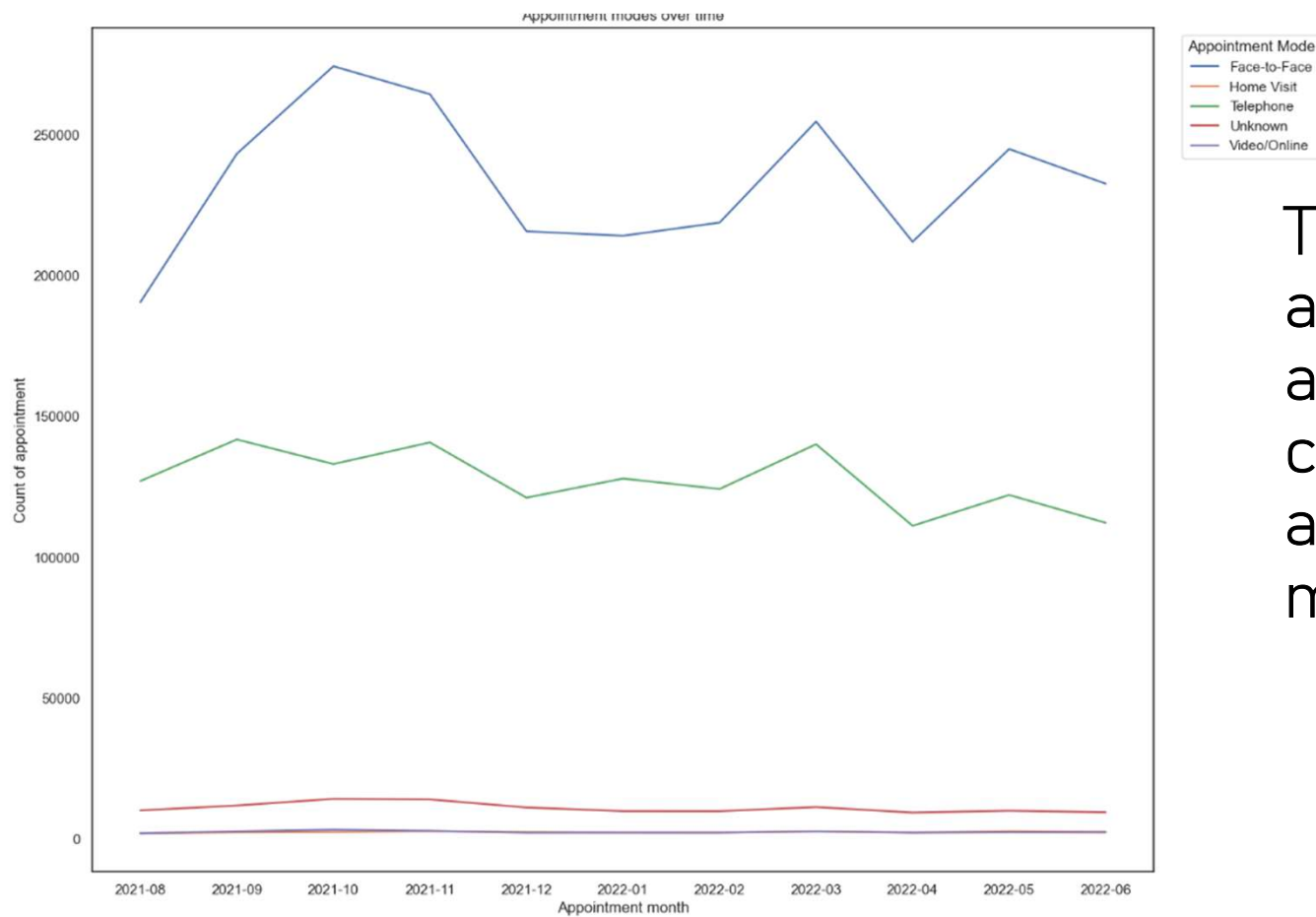




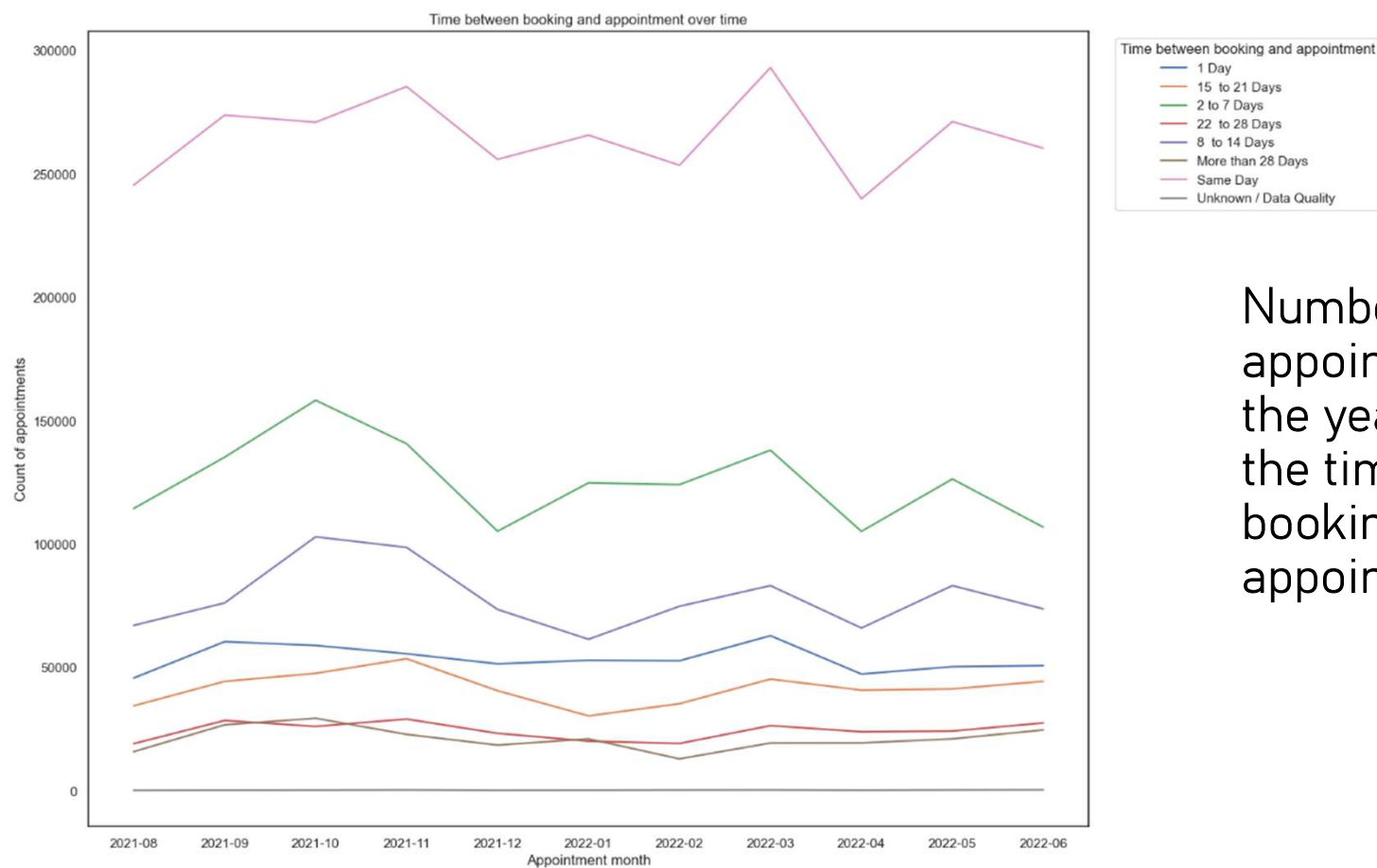
The number of appointments across the year for each healthcare professional



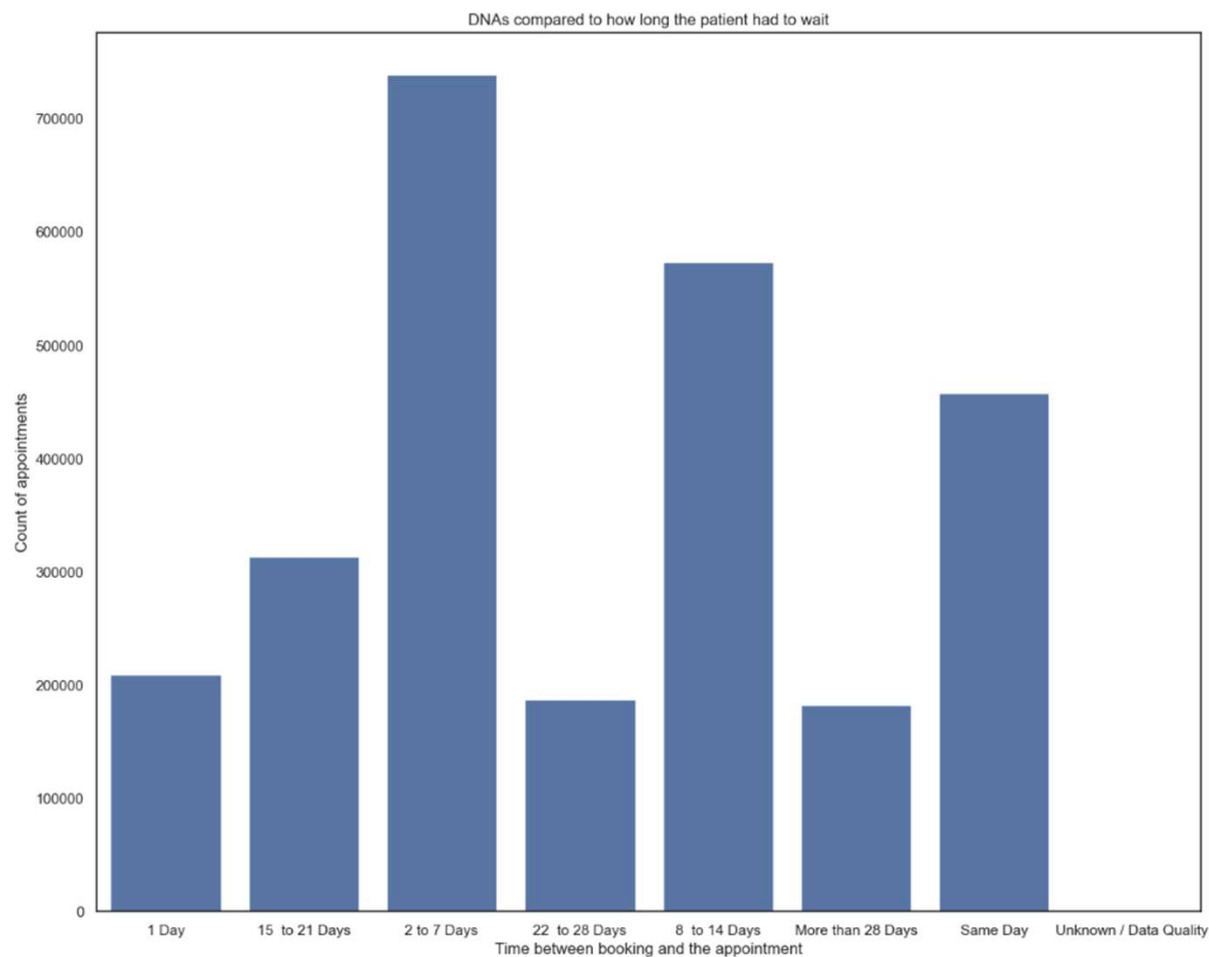
The number of appointments across the year compared to appointment status



The number of appointments across the year compared to appointment mode

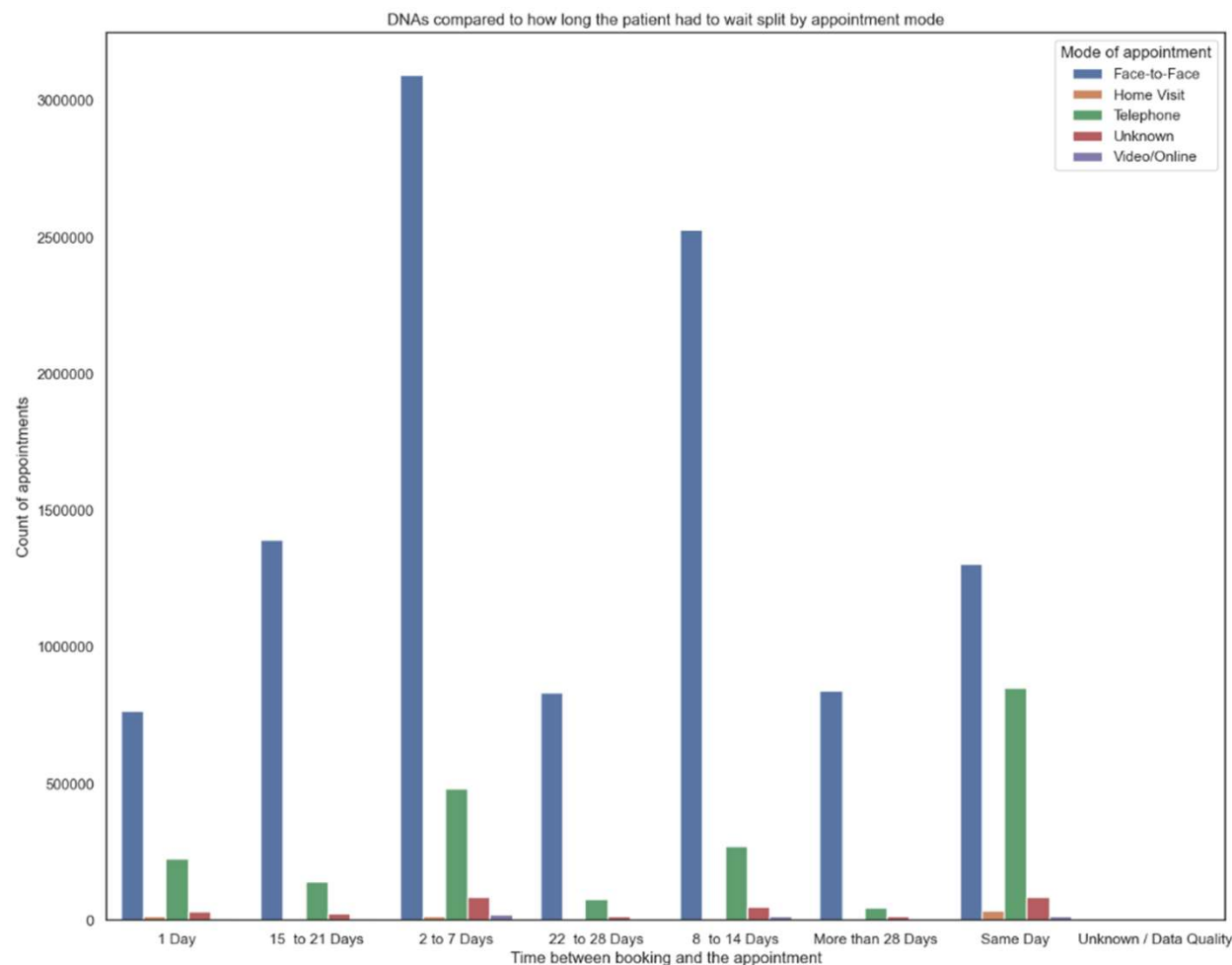


Number of appointments across the year compared to the time between booking and the appointment



The number of appointments that are missed compared to the time between booking and the appointment

The number of appointments missed compared to the time between booking and the appointment for each mode of appointment



Recommendations

- The NHS should increase the capacity to deal with increases in the population in the future.
 - Weekends and bank holidays
- As well as increasing capacity; there are also efficiencies that could be fixed.
 - Allowing GPs to do more telephone appointments
 - reminder messages can be sent out



A strategy

- Ultimately, it shows that a two-pronged approach towards the NHS capacity will increase the efficiency of the primary care system. By keeping the patients more up to date and having more specific times to avoid missed appointments, as well as increasing staffing levels and more appointments during weekends and bank holidays, there will be more room in this system to be used.



GEORGE BREESE

Thank you for
watching
